Arko Roy

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Technical Skills

Programming Languages: JavaScript, TypeScript, Solidity, Python, HTML, CSS, C++, C, Rust, Streamlit
Frameworks: React.js, Next.js, Node.js, TailwindCSS, Bootstrap, ShadCN, Aceternity,Motia
Databases: MongoDB, MySQL, PostgreSQL, Firebase, Prisma, Supabase
Developer Tools: Git, GitHub, Docker, VS Code, Foundry, Hardhat, Recoil, Zustand, Redis

Achievements

ETHGlobal New Delhi (2000+ participants) — Bounty Winner
Solana Frontier Hackathon 2026 — Bounty Winner
Vibethon 2026 — Winner
Build on CELO Kolkata Hackathon (100+ teams) — Winner
Avalanche Team1 Hackathon Kolkata (100+ teams) — Winner
Hack Heritage 2024 (1000+ participants) — Winner
EIBS IIT KGP — 2nd Place
EDU Chain Regional (100+ participants) — 3rd Place

Professional Experience

Blockchain Full Stack Developer — October 2025 – Current
Supermission

- Architected and integrated **autonomous AI agents using x402 and ERC-8004**, enabling agentic payments and multi-step task orchestration
- Implemented core product features including **prediction markets**, A2A payments, and fixing end-to-end user workflows

Blockchain Full Stack Developer — June 2025 – September 2025
VENQ

- Developed and deployed an ERC-1155 fractionalization smart contract, enabling real estate tokenization powering **10x growth** in asset handling using **Solidity**
- Revamped and migrated a legacy real estate investment platform, improving performance by **60%**, and modernizing the entire frontend with responsive design using **NextJS and Tailwind**

Frontend Developer Intern — April 2025 –June 2025
Research Commons

- Collaborated cross-functionally with backend and product teams to deliver responsive, modular UI components using **Next.js** and **Tailwind CSS**.

Project Experience

Envelope *Solana, Next.js, Anchor, Cloak, Encrypt, KIRAPAY, IKA* — **Solana Frontier 2026 Winner**

- Built a **private treasury and payroll operating system** for crypto-native teams, enabling FHE shielded salary disbursements, programmable treasury controls, and global contractor payments.
- Designed an end-to-end payroll orchestration system supporting **30+ contractors across 12+ countries**, with encrypted compensation policies, scoped auditing, and stealth payouts.

Arx Protocol *OP Stack, Solidity, Go, Foundry, Next.js, Docker* — **CELO Kolkata – Winner**

- Architected a **Layer-3 blockchain on OP Stack** featuring a novel **Proof-of-Impact consensus** mechanism for trust-minimized environmental verification.
- Implemented **ERC1155 Carbon Credits, Prediction Markets, and DAO governance**, alongside cross-chain settlement from **Celo → Ethereum**.

Verdict *Midnight, Compact, TypeScript, Next.js, SNARKs* — **Midnight Foundation Grant**

- Built a **Universal zero-knowledge integrity protocol** that verifies state transitions across applications without exposing private data with 10+ checks covering behavorial analysis and constraint
- Designed a verification engine with **10 layered integrity checks** covering sequence validity, behavioral analysis and constraint enforcement

Praxis *Solidity, Flare, Hardhat, Next.js, Wagmi, Noir, Groth16* — **Flare Foundation Grant**

- Designed a novel DeFi primitive called **Execution Rights**, allowing traders to execute strategies on vault capital without taking custody of assets.
- Built a reputation-driven capital allocation system with **ERC721 Execution Rights Tokens**, eliminating traditional lending defaults and liquidation risk.